

Operations on Singly Linked List

Q. Describe the fundamental operations performed on a Singly Linked List: Traversal, Insertion, Deletion, and Searching, detailing the logic behind each.

****Definition to Remember****

1. Traversal & Searching

Traversal (Printing the list):

Searching:

2. Insertion Operations

| Position | Logic Steps |

3. Deletion Operations

(Always check for Underflow: if `head == NULL`, list is empty).

| Position | Logic Steps |

4. Critical Edge Cases

* **Empty List:** Trying to delete causes underflow.

****Must-Write Points (for 10 marks)****

*****Quick Recall*****

*`Travers
free me*

Complete Executable C Code: Singly Linked List Full Implementation

```
#include <stdio.h>
```

#include

```
struct Node {
```

int data

```
// Insert at Head - O(1)
```

struct N

```
// Insert at Tail - O(n)
```

struct N

```
if (head == NULL) return newNode;
```

```
struct Node *temp = head;
```

while (t

```
// Delete by Value - O(n)
```

struct N

```
if (head->data == key) {
```

struct N

```
struct Node *curr = head;
```

```
while (curr != NULL) {
```

```
if (curr->next != NULL) {
```

```
struct Node *next = curr->next;
```

```
// Traverse List
```

```
void traverseList(struct Node *head) {
```

```
int main() {
```

```
struct Node *head = NULL;
```

```
head = insertAtHead(head, 20);
```

```
head = insertAtHead(head, 30);
```

```
printf("\n--- Deleting Node 20 ---\n");
```

```
head = deleteNode(head, 20);
```

```
return 0;
```

```
}
```